

A Colored Turán Proof of the Lollipop Formula

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Abstract

Let $a_{\mathcal{L}}(n)$ be the maximum number of connected regions into which n lollipops can divide the plane. Using Paulsen's two geometric obstructions and Karlsson's four-lollipop construction, we prove the exact four-cluster formula

$$a_{\mathcal{L}}(n) = 4 \binom{n}{2} + S(n) + n + 1,$$

where

$$S(n) = \max_{a+b+c+d=n, 0 \leq a \leq b \leq c \leq d} \left(3 \sum_{1 \leq i < j \leq 4} m_i m_j - 2ab \right),$$
$$(m_1, m_2, m_3, m_4) = (a, b, c, d).$$

After the geometric reduction, the upper bound is entirely combinatorial. It uses a colored Zykov symmetrization, a weighted blocker lemma, and an exact 3×4 matrix inequality. The matrix inequality is proved by support compression: alternating four-edge path compressions and double-star compressions reduce every integer matrix to star-forest normal form, where the efficient degree pattern is $K_{1,2} \cup K_2 \cup K_2$. This also explains the asymptotic cluster proportions

$$\frac{3}{16}, \frac{3}{16}, \frac{5}{16}, \frac{5}{16}.$$

1 The formula

A *lollipop* is a circle together with a half-line attached at one point of the circle and extending radially outward. We assume arrangements are generic: there are no tangencies, no triple intersection points, and no intersection occurs at an anchor point. This entails no loss for a maximum, since a non-generic arrangement can be perturbed without decreasing the number of regions.

Let C be the total number of crossings between distinct lollipops. Adding one generic lollipop with k crossings increases the number of regions by $k + 1$. Indeed, compactify the plane by a

point at infinity, so that all rays end at the same point. When the new lollipop is added, its k crossings subdivide k old edges and contribute k new vertices. The new anchor is one further vertex. The circle part contributes one more edge than the number of new vertices on the circle, and the ray part contributes one more edge than the number of new vertices on the ray. Thus $\Delta V = k + 1$ and $\Delta E = 2k + 2$, whence Euler's formula gives $\Delta F = \Delta E - \Delta V = k + 1$. Consequently

$$\#\text{regions} = C + n + 1. \quad (1)$$

Thus the problem is to maximize C .

For $n \geq 0$, define

$$S(n) = \max_{a+b+c+d=n, 0 \leq a \leq b \leq c \leq d} \left(3 \sum_{1 \leq i < j \leq 4} m_i m_j - 2ab \right), \quad (m_1, m_2, m_3, m_4) = (a, b, c, d). \quad (2)$$

Equivalently, for a fixed multiset of four cluster sizes, the penalized pair is the pair with smallest product, i.e. the two smallest clusters.

Theorem 1. *For every $n \geq 0$,*

$$a_{\mathcal{L}}(n) = 4 \binom{n}{2} + S(n) + n + 1. \quad (3)$$

The first values obtained from (3) are

$$\begin{aligned} &1, 2, 10, 25, 45, 71, 104, 142, 186, 237, 294, 356, 425, 500, 580, 667, \\ &761, 859, 964, 1076, 1193, 1316, 1446, 1581, 1722, 1870, 2024, 2183, \\ &2349, 2521, 2698, 2882, 3073. \end{aligned}$$

2 Geometric input and the two-graph reduction

We use Paulsen's terminology. A pair of lollipops is *close* if the angle between the two stem directions is at most 90° . A pair is *intriguing* if the two circles either do not meet, or meet at angle at most 90° , after orienting the two circles in the same direction.

The following geometric facts are due to Cutler–Karlsson–Sloane and Paulsen. The present paper proves the exact combinatorial extremal step that follows from these facts.

Lemma 2 (geometric lollipop input). *For a generic pair of lollipops:*

- (i) *any pair has at most 7 crossings;*
- (ii) *a close pair has at most 5 crossings;*
- (iii) *an intriguing pair has at most 5 crossings;*
- (iv) *a pair that is both close and intriguing has at most 4 crossings.*

Furthermore, among any four lollipops there is a close pair, and among any five lollipops there is an intriguing pair.

For (i), two circles meet at most twice, each stem meets the other circle at most twice, and the two stems meet at most once. The remaining assertions are the geometric obstructions quoted above; Appendix A recalls Paulsen's linear-algebra proof of the forced intriguing pair.

Given an arrangement of n lollipops, define graphs D, E on the same vertex set by

$$\begin{aligned} ij \in D &\iff L_i, L_j \text{ are not close,} \\ ij \in E &\iff L_i, L_j \text{ are not intriguing.} \end{aligned}$$

By Lemma 2,

$$D \text{ is } K_4\text{-free,} \quad E \text{ is } K_5\text{-free.} \quad (4)$$

For a pair ij , Lemma 2 gives the pointwise estimate

$$c_{ij} \leq 4 + \mathbf{1}_{ij \in D} + \mathbf{1}_{ij \in E} + \mathbf{1}_{ij \in D \cap E}. \quad (5)$$

Indeed, the four cases are respectively: close and intriguing, close only, intriguing only, and neither; the corresponding upper bounds are 4, 5, 5, 7. Therefore

$$C \leq 4 \binom{n}{2} + \sigma(D, E), \quad \sigma(D, E) := |D| + |E| + |D \cap E|. \quad (6)$$

Theorem 1 is now reduced to the following colored Turán theorem.

Theorem 3 (colored Turán theorem). *If D is K_4 -free and E is K_5 -free on the same n -vertex set, then*

$$\sigma(D, E) \leq S(n). \quad (7)$$

The proof occupies Sections 3–5. The lower construction is given in Section 6; together with (1) and (6), it proves Theorem 1.

3 Colored Zykov symmetrization

Color every unordered pair of vertices with one of four colors:

$$\begin{aligned} 0 &: \text{neither } D \text{ nor } E, & A &: E\text{-only,} \\ B &: D\text{-only,} & X &: D \cap E. \end{aligned}$$

These colors have weights

$$w(0) = 0, \quad w(A) = 1, \quad w(B) = 1, \quad w(X) = 3,$$

so that $\sigma(D, E)$ is the total color-weight.

Lemma 4 (colored Zykov reduction). *An extremal pair (D, E) for σ may be chosen as a blow-up of a complete colored quotient: the vertex set is partitioned into classes $V_1, \dots, V_k$, every pair inside one class has color 0, and every pair between two distinct classes has one fixed color from $\{A, B, X\}$. In particular, there are no color-0 pairs between distinct classes.*

Proof. Call two vertices *zero-twins* if either they are equal, or their pair has color 0 and they have the same colored neighborhood outside the pair. This is an equivalence relation. Among all feasible colorings attaining the maximum value of σ , choose one maximizing the sum of squares of the zero-twin class sizes.

For a zero-twin class P , let $d_w(P)$ be the total color-weight from any vertex of P to vertices outside the two classes under comparison; this is well-defined because vertices in P are twins. Suppose two distinct zero-twin classes P, Q have a color-0 edge between them. Then all edges between P and Q have color 0. If $d_w(P) \leq d_w(Q)$, clone every vertex of P to a representative of Q : keep all pairs inside $P \cup Q$ color 0, and give each vertex of P the same colors to the outside as a vertex of Q . The change in σ is $|P|(d_w(Q) - d_w(P)) \geq 0$.

This operation cannot create a K_4 in D . Indeed, a new D -clique using a cloned vertex from P and no vertex of Q corresponds to the old clique obtained by replacing that vertex by a representative of Q , while a clique using vertices from both P and Q is impossible because all pairs inside $P \cup Q$ have color 0. The same argument forbids creation of a K_5 in E . Thus feasibility is preserved. If $d_w(P) < d_w(Q)$, this contradicts extremality of σ ; if $d_w(P) = d_w(Q)$, it preserves σ but merges P and Q into one larger zero-twin class, contradicting the tie-breaker. The case $d_w(Q) < d_w(P)$ is symmetric.

Hence no color-0 edge joins two distinct zero-twin classes. Between two classes all edges have one fixed nonzero color, because vertices within each class have identical colored neighborhoods. This is precisely the claimed quotient form. $\square$

Let the quotient class sizes be $x_1, \dots, x_k$, with $\sum_i x_i = n$, and put

$$Q = \sum_i x_i^2.$$

Let

$$a = \sum_{ij \in A} x_i x_j, \quad b = \sum_{ij \in B} x_i x_j,$$

where A and B now also denote the graphs of A -colored and B -colored quotient edges. The total cross-class mass is

$$P = \sum_{i < j} x_i x_j = \frac{n^2 - Q}{2}.$$

Since X -edges have weight 3, while A - and B -edges have weight 1,

$$\sigma = 3P - 2a - 2b = \frac{3}{2}(n^2 - Q) - 2a - 2b. \quad (8)$$

The forbidden cliques translate into blocker conditions. Since D consists of colors $B \cup X$, every four quotient vertices contain an A -edge; otherwise they form a K_4 in D . Thus

$$\alpha(A) \leq 3. \quad (9)$$

Similarly, since E consists of colors $A \cup X$,

$$\alpha(B) \leq 4. \quad (10)$$

4 The weighted blocker lemma

For nonnegative integer weights $x_1, \dots, x_k$ with total n , define $Q = \sum_i x_i^2$. If G is a graph on $[k]$, write

$$w(G) = \sum_{ij \in E(G)} x_i x_j.$$

For $r \geq 1$, let

$$\rho_r(x) = \min_P \sum_{P \in \mathcal{P}} \left(\sum_{i \in P} x_i \right)^2,$$

where the minimum is over all partitions of $[k]$ into r possibly empty parts.

Lemma 5 (weighted Turán theorem). *If G is K_{r+1} -free, then*

$$w(G) \leq \frac{1}{2}(n^2 - \rho_r(x)). \quad (11)$$

Proof. The usual Zykov symmetrization works with vertex weights. If two nonadjacent vertices u, v have weighted neighborhoods $d_w(u) = \sum_{z \sim u} x_z$ and $d_w(v) = \sum_{z \sim v} x_z$, and $d_w(u) \leq d_w(v)$, then replacing u by a clone of v changes $w(G)$ by $x_u(d_w(v) - d_w(u)) \geq 0$. It also preserves K_{r+1} -freeness, for the same reason as in Lemma 4. After the standard finite tie-breaking, an extremal weighted graph is complete multipartite. Since it is K_{r+1} -free, it has at most r parts. For a partition with part weights $s_1, \dots, s_r$, its edge weight is

$$\sum_{i < j} s_i s_j = \frac{1}{2} \left(n^2 - \sum_i s_i^2 \right),$$

which is maximized exactly when $\sum_i s_i^2$ is minimized. □

Define

$$M(n) = \min_{a+b+c+d=n, a,b,c,d \in \mathbb{Z}_{\geq 0}} \left[\frac{3}{2}(a^2 + b^2 + c^2 + d^2) + 2ab \right]. \quad (12)$$

Since

$$\frac{3}{2}n^2 - \left[\frac{3}{2}(a^2 + b^2 + c^2 + d^2) + 2ab \right] = 3 \sum_{i < j} m_i m_j - 2ab,$$

where $(m_1, m_2, m_3, m_4) = (a, b, c, d)$, and since the two variables in the penalized product may be labelled as the two smallest cluster sizes, the definitions imply

$$S(n) = \frac{3}{2}n^2 - M(n). \quad (13)$$

Lemma 6 (weighted blocker lemma). *Let A, B be graphs on $[k]$ with $\alpha(A) \leq 3$ and $\alpha(B) \leq 4$. With the notation above,*

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq M(n). \quad (14)$$

Proof. The complement of A is K_4 -free. By Lemma 5,

$$\frac{1}{2}(n^2 - Q) - w(A) \leq \frac{1}{2}(n^2 - \rho_3(x)),$$

and hence

$$w(A) \geq \frac{1}{2}(\rho_3(x) - Q). \quad (15)$$

Likewise, the complement of B is K_5 -free, so

$$w(B) \geq \frac{1}{2}(\rho_4(x) - Q). \quad (16)$$

Combining (15) and (16),

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq \rho_3(x) + \rho_4(x) - \frac{1}{2}Q. \quad (17)$$

Choose partitions attaining $\rho_3(x)$ and $\rho_4(x)$, and intersect them. This gives a 3×4 nonnegative integer matrix $U = (u_{ij})$, where u_{ij} is the total weight lying in the i -th part of the 3-partition and the j -th part of the 4-partition. Let

$$r_i = \sum_j u_{ij}, \quad c_j = \sum_i u_{ij}.$$

Then

$$\rho_3(x) = \sum_i r_i^2, \quad \rho_4(x) = \sum_j c_j^2.$$

Also

$$Q = \sum_\ell x_\ell^2 \leq \sum_{i,j} u_{ij}^2,$$

because squaring after grouping can only increase the sum of squares. Thus (17) gives

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq \sum_i r_i^2 + \sum_j c_j^2 - \frac{1}{2} \sum_{i,j} u_{ij}^2.$$

The right side is at least $M(n)$ by the matrix theorem proved in Section 5. □

We now prove Theorem 3, assuming the matrix theorem for a moment.

Proof of Theorem 3. By Lemma 4, assume (D, E) is in quotient form. Then (9) and (10) allow Lemma 6, so

$$\frac{3}{2}Q + 2a + 2b \geq M(n).$$

Using (8),

$$\sigma = \frac{3}{2}n^2 - \left(\frac{3}{2}Q + 2a + 2b \right) \leq \frac{3}{2}n^2 - M(n) = S(n),$$

by (13). This proves (7). □

5 The 3×4 matrix theorem

For a 3×4 nonnegative integer matrix $U = (u_{ij})$ with total sum n , row sums r_i , and column sums c_j , put

$$F(U) = \sum_{i=1}^3 r_i^2 + \sum_{j=1}^4 c_j^2 - \frac{1}{2} \sum_{i,j} u_{ij}^2. \quad (18)$$

Theorem 7 (matrix theorem). *For every 3×4 nonnegative integer matrix U of total sum n ,*

$$F(U) \geq M(n). \quad (19)$$

The proof is the main technical point. Think of the positive entries of U as edge masses on the bipartite graph $K_{3,4}$, whose left vertices are the rows and whose right vertices are the columns. If two positive cells share a row or a column, call the corresponding edges adjacent. Expanding (18) gives

$$F(U) = \frac{3}{2} \sum_e u_e^2 + 2 \sum_{e \sim f} u_e u_f. \quad (20)$$

Compression to a star forest

The next lemma is the normal-form step needed below: every matrix can be compressed to a star forest without increasing F . Equality may occur during compression, so the conclusion concerns the compressed matrix rather than the original support.

Lemma 8 (support compression). *For every nonnegative integer matrix U , there is a nonnegative integer matrix U' with the same total sum such that*

$$F(U') \leq F(U)$$

and the support graph of U' is a forest whose connected components are stars.

Proof. First remove cycles. If the support contains a cycle, add ε and $-\varepsilon$ alternately around the cycle. Row sums and column sums, and therefore the first two terms of (18), remain fixed. The final term $-\frac{1}{2} \sum u_{ij}^2$ is a concave quadratic in ε , so one endpoint of the maximal integer interval of allowed ε 's does not increase F and makes at least one entry zero. Repeating leaves a forest.

Now suppose a tree component has diameter at least 4. Choose four consecutive edges on a path in that component, with masses x_1, x_2, x_3, x_4 . Replace them by

$$x_1 + \varepsilon, \quad x_2 - \varepsilon, \quad x_3 + \varepsilon, \quad x_4 - \varepsilon.$$

The total mass is preserved. The three internal vertex sums on this four-edge path are unchanged; only the two endpoint vertex sums change. In the expression (20), the coefficient of ε^2 is

$$2 - \frac{1}{2} \cdot 4 = 0,$$

so F is linear in ε on the allowed integer interval. One endpoint of that interval does not increase F and kills at least one of the four entries. Repeating, no component of diameter at least 4 remains.

It remains to treat a component of diameter exactly 3, i.e. a double-star. Let y be the mass of the central edge. Choose a leaf edge of mass x on one side and a leaf edge of mass z on the other. Let A be the sum of the other leaf-edge masses incident with the center on the x -side, and let B be the analogous sum on the z -side. If we delete the central edge and add its mass y to the x -edge, a direct calculation from (20) gives the change

$$F_{\text{old}} - F_x = y(2B + 2z - x).$$

If instead we add y to the z -edge, then

$$F_{\text{old}} - F_z = y(2A + 2x - z).$$

At least one of these two quantities is nonnegative: otherwise $x > 2B + 2z \geq 2z$ and $z > 2A + 2x \geq 2x$, impossible. Hence one of the two operations preserves the total mass, keeps all masses integral and nonnegative, deletes the central edge, and does not increase F . Repeating this step removes all diameter-3 components.

A forest whose components have diameter at most 2 is precisely a star forest. This proves the lemma. $\square$

Star forests

For a star component with t positive edges of masses $z_1, \dots, z_t$ and total mass $s = \sum_{\ell} z_{\ell}$, its contribution to F is

$$s^2 + \frac{1}{2} \sum_{\ell=1}^t z_{\ell}^2. \quad (21)$$

In particular,

$$s^2 + \frac{1}{2} \sum_{\ell=1}^t z_{\ell}^2 \geq \left(1 + \frac{1}{2t}\right) s^2 = \frac{s^2}{\eta_t}, \quad \eta_t := \frac{2t}{2t+1}. \quad (22)$$

Lemma 9 (star-forest minimum). *If U is a nonnegative integer matrix whose support is a star forest, then $F(U) \geq M(n)$, where n is the total sum of its entries.*

Proof. If $n = 0$, then $F(U) = M(0) = 0$. Hence assume $n > 0$. Let the star components have degrees t_{ν} and total masses s_{ν} . If the degree list is not $(2, 1, 1)$, then

$$\Gamma := \sum_{\nu} \eta_{t_{\nu}} \leq 2. \quad (23)$$

Indeed, a star forest in $K_{3,4}$ has at most three nontrivial components, because each such component uses at least one of the three row vertices. One or two components give $\Gamma < 2$, since each $\eta_t < 1$. If there are three components, then the seven vertices of $K_{3,4}$ give $\sum_{\nu} (t_{\nu} + 1) \leq 7$, so the only possible degree lists are $(2, 1, 1)$ and $(1, 1, 1)$. Excluding $(2, 1, 1)$ leaves $(1, 1, 1)$, for which $\Gamma = 3 \cdot (2/3) = 2$.

By (22) and Cauchy's inequality,

$$F(U) \geq \sum_{\nu} \frac{s_{\nu}^2}{\eta_{t_{\nu}}} \geq \frac{(\sum_{\nu} s_{\nu})^2}{\sum_{\nu} \eta_{t_{\nu}}} = \frac{n^2}{\Gamma} \geq \frac{n^2}{2}.$$

For $n \geq 4$, the balanced quadruple in (12) gives $M(n) \leq n^2/2$. Explicitly, for $n = 4q + r$, use (q, q, q, q) , $(q, q, q, q + 1)$, $(q, q, q + 1, q + 1)$, or $(q, q + 1, q + 1, q + 1)$ according as $r = 0, 1, 2, 3$. The resulting value is at most $n^2/2$ for $n \geq 4$. For $n \leq 3$, the identity

$$F(U) = \frac{3}{2}n + 3 \sum_{i,j} \binom{u_{ij}}{2} + 2 \sum_{\substack{(i,j) < (i',j') \\ (i=i' \text{ or } j=j'), (i,j) \neq (i',j')}} u_{ij} u_{i'j'}$$

shows $F(U) \geq \frac{3}{2}n = M(n)$. Thus every non- $(2, 1, 1)$ star forest satisfies the desired bound.

Now suppose the degree list is $(2, 1, 1)$. The degree-two star must be centered at a row vertex: if it were centered at a column vertex, only one row vertex would remain for the two disjoint K_2 components. Thus, up to row and column permutation, its four positive edge masses are

$$\begin{pmatrix} a & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & d \end{pmatrix}.$$

For this matrix,

$$F(U) = (a + b)^2 + c^2 + d^2 + \frac{1}{2}(a^2 + b^2 + c^2 + d^2) = \frac{3}{2}(a^2 + b^2 + c^2 + d^2) + 2ab.$$

Since $a + b + c + d = n$, this is at least $M(n)$ by definition. $\square$

Proof of Theorem 7. Apply Lemma 8 to replace U by a star-forest matrix U' with the same total sum and $F(U') \leq F(U)$. Lemma 9 gives $F(U') \geq M(n)$. Hence $F(U) \geq M(n)$. $\square$

6 The lower construction

The lower construction is a blow-up of Karlsson's optimal four-lollipop configuration. We record explicit parameters to make the construction reproducible. Write $Q_{\text{anch}}(x, y, r, \theta)$ for the lollipop with anchor (x, y) , radius r , and stem direction $(\cos \theta, \sin \theta)$; its center is $(x, y) - r(\cos \theta, \sin \theta)$. Write $Q_{\text{cent}}(x, y, r, \theta)$ for the same object specified by center (x, y) . Let

$$\begin{aligned} L_1 &= Q_{\text{anch}}(0, 0, 200, 0), \\ L_2 &= Q_{\text{cent}}(0.45, 0.4, 0.55, -0.01), \\ L_3 &= Q_{\text{anch}}\left(1.15, 0.65, 100, -\frac{\pi}{2} - \frac{2\pi}{15}\right), \\ L_4 &= Q_{\text{anch}}\left(1.0488116827495215, -0.05, 100, \frac{\pi}{2} + \frac{\pi}{30}\right). \end{aligned}$$

These coordinates are Karlsson's four-lollipop arrangement, as listed with the OEIS entry A389624. Solving the elementary pairwise equations for circle-circle, circle-ray, and ray-ray

intersections gives the crossing table

	1	2	3	4
1	*	5	7	7
2		*	7	7
3			*	7
4				*

and all listed intersections are transverse, away from anchors, and away from parallel stem degeneracies. Hence the table is stable under sufficiently small independent perturbations of the four lollipops.

Lemma 10 (blow-up stability). *For any nonnegative integers m_1, m_2, m_3, m_4 , there is a generic arrangement of $m_1 + m_2 + m_3 + m_4$ lollipops whose crossing number is*

$$L(m_1, m_2, m_3, m_4) = 4 \sum_i \binom{m_i}{2} + 5m_1m_2 + 7 \sum_{\substack{i < j \\ (i,j) \neq (1,2)}} m_i m_j. \quad (24)$$

Proof. Because the base configuration above is generic and all stem directions in distinct base pairs are nonparallel, there are parameter neighborhoods $\mathcal{U}_i$ of L_i such that any lollipop in $\mathcal{U}_i$ and any lollipop in $\mathcal{U}_j$, $i \neq j$, have the same number of crossings as L_i and L_j . This is immediate from the explicit intersection equations: circle–circle intersections are governed by strict inequalities between center distance and sums or differences of radii, circle–ray intersections by a quadratic whose accepted roots have ray parameter strictly positive, and ray–ray intersections by a nonsingular linear system whose ray parameters are either strictly positive or strictly outside the ray.

It remains to choose, inside each $\mathcal{U}_i$, as many copies as desired so that every intra-cluster pair has exactly four crossings. By applying a similarity, it suffices to do this for the unit lollipop Λ with center $(0,0)$, radius 1, and stem $\{(1+s, 0) : s \geq 0\}$. Compare Λ with the lollipop Λ_t of radius 1, center (t,t) , and stem direction angle $-t$. For all sufficiently small $t > 0$, the two circles meet twice. The stem of Λ meets the circle of Λ_t exactly once, since the intersections with the line $y = 0$ have $x = t \pm \sqrt{1-t^2}$, and exactly the plus root is greater than 1. The stem of Λ_t meets the circle of Λ in no ray point: the positive line-intersection root is $1 - t + O(t^2) < 1$, while the other root is negative. Finally, the two stems meet once, because the stem of Λ_t has the form $(t, t) + q(\cos t, -\sin t)$, $q \geq 1$, and meets the x -axis at $q = t/\sin t > 1$ and $x = 1 + t + O(t^2) > 1$. These four crossings are transverse.

The preceding inequalities are strict, so the same four-crossing conclusion holds throughout a small open chamber in the relative local parameter space. To obtain arbitrarily many pairwise four-crossing copies, choose a small one-parameter motion contained in this chamber. For instance, if R_α denotes rotation by α , set

$$\Phi_t(x) = R_{-t}x + \int_0^t R_{-\tau}(1, 1) d\tau.$$

Then $\Phi_{s+t} = \Phi_s \circ \Phi_t$, and $\Phi_t\Lambda$ has center $(t, t) + O(t^2)$ and stem direction angle $-t$. Hence, for all sufficiently small $t > 0$, the pair $\Lambda, \Phi_t\Lambda$ lies in the same four-crossing chamber as the model pair Λ, Λ_t . It follows from the group law that any two distinct members $\Phi_s\Lambda$ and $\Phi_t\Lambda$,

with $0 < s < t$ sufficiently small, have the same crossing number as $\Lambda, \Phi_{t-s}\Lambda$, namely four.

Applying a similarity to this local family at each base lollipop and taking the parameters sufficiently small, we may choose m_i distinct copies inside $\mathcal{U}_i$. Then every pair inside the i -th cluster has exactly four crossings, while every pair from two different clusters has the base crossing number. A final arbitrarily small perturbation, still within the same open chambers, removes accidental triple intersection points and anchor incidences without changing any pairwise crossing count. This gives (24). $\square$

Subtracting $4\binom{n}{2}$ from (24), with $n = m_1 + m_2 + m_3 + m_4$, gives

$$L(m_1, m_2, m_3, m_4) - 4\binom{n}{2} = 3 \sum_{i < j} m_i m_j - 2m_1 m_2.$$

For a fixed multiset of four cluster sizes, the product penalized by the coefficient -2 should be the smallest product. Hence the best blow-up has

$$\max L(m_1, m_2, m_3, m_4) = 4\binom{n}{2} + S(n).$$

By (1), this gives

$$a_{\mathcal{L}}(n) \geq 4\binom{n}{2} + S(n) + n + 1. \quad (25)$$

The upper bound follows from Lemma 2, Theorem 3, and (6). Thus (25) is sharp, proving Theorem 1.

7 Numerical table and asymptotics

The continuous relaxation of (12) is minimized at

$$a = b = \frac{3n}{16}, \quad c = d = \frac{5n}{16}.$$

Consequently

$$S(n) = \frac{33}{32}n^2 + O(1),$$

and

$$a_{\mathcal{L}}(n) = 4\binom{n}{2} + \frac{33}{32}n^2 + n + O(1) = \frac{97}{32}n^2 - n + O(1).$$

n	one optimal (a, b, c, d)	$S(n)$	$a_{\mathcal{L}}(n)$
0	(0,0,0,0)	0	1
1	(0,0,0,1)	0	2
2	(0,0,1,1)	3	10
3	(0,1,1,1)	9	25
4	(1,1,1,1)	16	45
5	(1,1,1,2)	25	71
6	(1,1,2,2)	37	104
7	(1,2,2,2)	50	142
8	(1,2,2,3)	65	186
9	(1,2,3,3)	83	237
10	(2,2,3,3)	103	294
11	(2,2,3,4)	124	356
12	(2,2,4,4)	148	425
13	(2,3,4,4)	174	500
14	(2,3,4,5)	201	580
15	(2,3,5,5)	231	667
16	(3,3,5,5)	264	761
17	(3,3,5,6)	297	859
18	(3,3,6,6)	333	964
19	(3,4,6,6)	372	1076
20	(4,4,6,6)	412	1193
21	(4,4,6,7)	454	1316
22	(4,4,7,7)	499	1446
23	(4,5,7,7)	545	1581
24	(4,5,7,8)	593	1722
25	(4,5,8,8)	644	1870
26	(5,5,8,8)	697	2024
27	(5,5,8,9)	751	2183
28	(5,5,9,9)	808	2349
29	(5,6,9,9)	867	2521
30	(5,6,9,10)	927	2698

Table 1: Values from the closed formula.

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A Paulsen's forced intriguing pair

We recall the short linear-algebra proof that among five circles some pair is intriguing. Suppose five circles with centers $x_i \in \mathbb{R}^2$ and radii $r_i > 0$ are pairwise non-intriguing. Then every pair intersects at an obtuse angle, equivalently

$$r_i^2 + r_j^2 < |x_i - x_j|^2 < (r_i + r_j)^2. \quad (26)$$

For each circle set

$$v_i = \frac{1}{r_i}(1, r_i^2 - |x_i|^2, x_i) \in \mathbb{R}^4,$$

and endow $\mathbb{R}^4$, written as triples (α, β, x) , with the symmetric bilinear form

$$\langle (\alpha, \beta, x), (\alpha', \beta', x') \rangle = \frac{1}{2}\alpha\beta' + \frac{1}{2}\beta\alpha' + x \cdot x'.$$

A direct calculation using (26) gives

$$\langle v_i, v_i \rangle = 1, \quad -1 < \langle v_i, v_j \rangle < 0 \quad (i \neq j).$$

Five vectors in $\mathbb{R}^4$ are linearly dependent. In a nontrivial relation $\sum_i \lambda_i v_i = 0$, the coefficients cannot all have the same sign, because the first coordinate of every v_i is $1/r_i > 0$. Therefore we may move the negative terms to the other side and write

$$\sum_{i \in P} a_i v_i = \sum_{j \in N} b_j v_j, \quad a_i, b_j > 0,$$

with P, N nonempty and disjoint. Replacing the relation by its negative if necessary, assume $|P| \leq 2$. Put $w = \sum_{i \in P} v_i$. If $|P| = 1$, then $\langle \sum_{i \in P} a_i v_i, w \rangle > 0$. If $P = \{1, 2\}$, it equals $(a_1 + a_2)(1 + \langle v_1, v_2 \rangle) > 0$. But

$$\left\langle \sum_{j \in N} b_j v_j, w \right\rangle < 0,$$

since all cross inner products are negative. This contradicts the linear relation. Therefore five pairwise non-intriguing circles cannot exist.